

StudentWise — Full Project Specification

Document purpose:

Project:

Repository:

Last updated:

Presentation-ready spec for PPT slides

StudentWise University Portal Dashboard

github.com/mangiai/studentwise-dash

June 10, 2026

1. Executive Summary

StudentWise is a full-stack university portal that lets students, teachers, and administrators manage courses, attendance, fees, grades, and notifications from a single dashboard. The app is built with **React 19 + TanStack Start**, deployed on **Vercel**, and backed by **Supabase Postgres** with Row Level Security (RLS).

Attribute	Value
Product name	StudentWise
Tagline	University Portal
Package name	studentwise-dash
Local dev URL	http://localhost:3000
Production URL	https://studentwise-dash.vercel.app <i>(configure in Vercel)</i>
GitHub	https://github.com/mangiai/studentwise-dash
Source files	~86 TypeScript/TSX files (~9,400 lines)
Total commits	11 (single-day sprint: June 10, 2026)

2. Problem Statement & Goals

Problem

Universities need a centralized digital portal where students can track academic progress, fees, and attendance; teachers can view analytics; and admins can manage enrollment data — without juggling multiple disconnected systems.

Goals Achieved

- Role-based portal for **students**, **teachers**, and **admins**
- Live data from Supabase (not static mock data)
- Secure authentication with session cookies (SSR-compatible)
- Admin CRUD for students, teachers, courses, and enrollments
- Production deployment on Vercel with Nitro serverless functions
- Demo-ready seed data and test accounts for presentations

3. Technology Stack

Frontend

Technology	Version	Purpose
React	19.2	UI framework
TanStack Router	1.168	File-based routing
TanStack Start	1.167	Full-stack React framework (SSR + server functions)
TanStack React Query	5.83	Server state / data fetching
Tailwind CSS	4.2	Utility-first styling
shadcn/ui + Radix UI	Latest	Accessible component library (40+ UI primitives)
Recharts	2.15	Dashboard charts and analytics
Lucide React	0.575	Icon set
React Hook Form + Zod	7.71 / 3.24	Form validation
Sonner	2.0	Toast notifications
date-fns	4.1	Date formatting

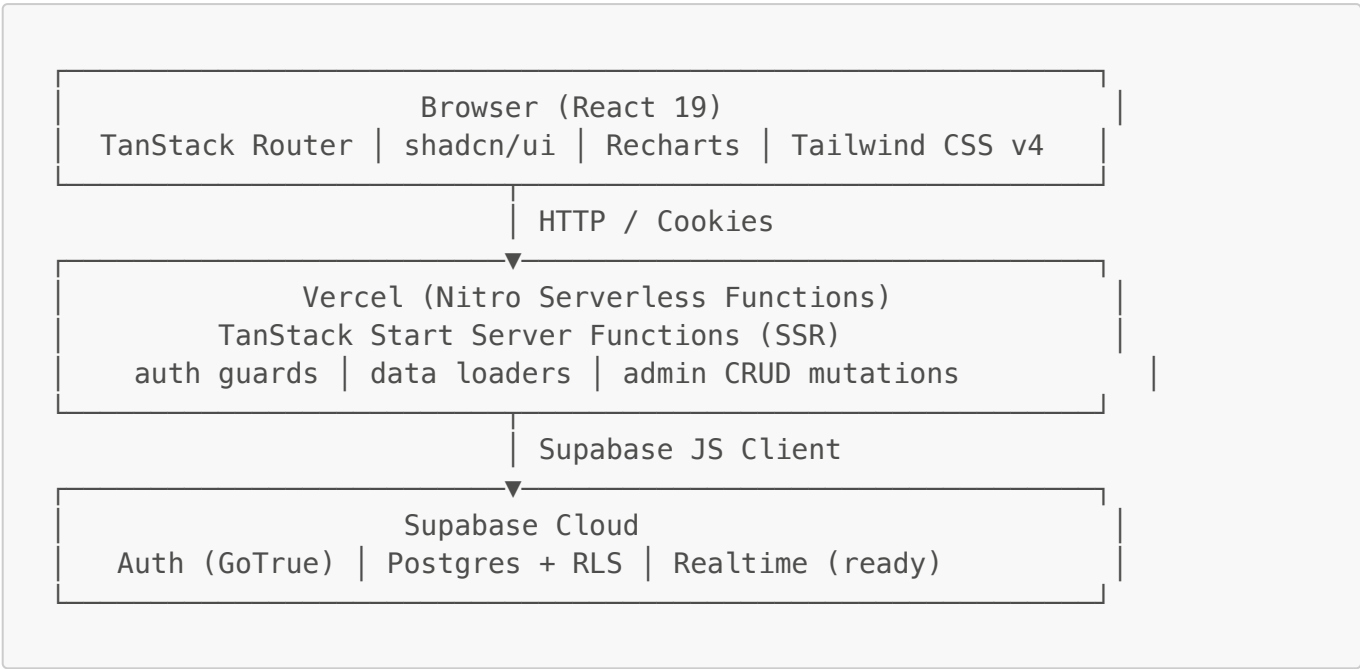
Backend & Database

Technology	Purpose
Supabase Postgres	Primary database
Supabase Auth	Email/password authentication
@supabase/ssr	Cookie-based sessions for SSR
Row Level Security (RLS)	Per-role data access policies
pgcrypto	Password hashing for seed users

Build, Deploy & Tooling

Technology	Purpose
Vite	7.3 — Dev server and bundler
Nitro	3.0 — Serverless preset for Vercel
Node.js	22.x — Runtime (local + Vercel functions)
TypeScript	5.8 — Type safety
ESLint + Prettier	Code quality
Supabase CLI	Migrations, local DB, type generation

Architecture Diagram



4. User Roles & Access Control

Role	Login URL	Access
Student	/login	Dashboard, courses, attendance, fees, teachers, results, notifications, settings
Teacher	/login	All student pages + Reports & Analytics
Admin	/admin/login	Full portal + Admin Dashboard (CRUD)

Auth Guards

- `requireAuth` — redirects unauthenticated users to `/login`
- `requireGuest` — redirects signed-in users away from login/signup
- `requireAdmin` — restricts admin routes to `role = admin`
- `requireAdminGuest` — admin login page guard

Session Model

- Supabase Auth with **HTTP cookies** via `@supabase/ssr`
- Profile auto-created on signup via Postgres trigger (`handle_new_user`)
- Role stored in `auth.users.raw_app_meta_data.role` and `profiles.role`

5. Feature Map — All Routes & Links

Replace `{BASE_URL}` with `http://localhost:3000` (local) or your Vercel production URL.

5.1 Public / Auth Routes

Feature	Route	Link	Description
Home redirect	/	{BASE_URL}/	Redirects to /dashboard if logged in, else /login
Student/Teacher Login	/login	{BASE_URL}/login	Email + password sign-in
Admin Login	/admin/login	{BASE_URL}/admin/login	Separate admin portal entry
Sign Up	/signup	{BASE_URL}/signup	New student registration + student ID linking

5.2 Student & Shared Portal Routes

Feature	Route	Link	Roles	Live Data
Dashboard	/dashboard	{BASE_URL}/dashboard	All	☒ Stats, charts, GPA, attendance overview
My Courses	/courses	{BASE_URL}/courses	All	☒ Enrollments, credits, attendance badges
Attendance	/attendance	{BASE_URL}/attendance	All	☒ Per-course %, ring charts, short-attendance alerts
Fee Management	/fees	{BASE_URL}/fees	Student, Admin	☒ Semester balance, payment history (PKR)
Faculty Directory	/teachers	{BASE_URL}/teachers	All	☒ Teacher cards from database
Academic Results	/results	{BASE_URL}/results	All	☒ Semester grades, CGPA, grade badges
Notifications	/notifications	{BASE_URL}/notifications	All	☒ Fee, attendance, course, announcement alerts
Settings	/settings	{BASE_URL}/settings	All	Profile, notification preferences (UI)

5.3 Teacher & Admin Analytics

Feature	Route	Link	Roles	Live Data
Reports & Analytics	/reports	{BASE_URL}/reports	Teacher, Admin	✔ Enrollment trends, fee status, attendance, department charts

5.4 Admin Panel

Feature	Route	Link	Roles	Live Data
Admin Dashboard	/admin/dashboard	{BASE_URL}/admin/dashboard	Admin only	✔ Full CRUD operations

Admin capabilities:

- Create / edit / delete **students**
- Create / edit / delete **teachers**
- Create / edit / delete **courses**
- **Enroll / unenroll** students in courses
- **Assign instructors** to courses
- Search and filter across all entities
- Tabbed interface: Students | Teachers | Courses

5.5 Sidebar Navigation (Role-Filtered)

Defined in `AppLayout.tsx` — each link appears only for allowed roles:

Nav Item	Route	Student	Teacher	Admin
Dashboard	/dashboard	✔	✔	✔
My Courses	/courses	✔	✔	✔
Attendance	/attendance	✔	✔	✔
Fee Management	/fees	✔	—	✔
Teachers	/teachers	✔	✔	✔
Results	/results	✔	✔	✔
Reports	/reports	—	✔	✔
Admin Panel	/admin/dashboard	—	—	✔
Notifications	/notifications	✔	✔	✔
Settings	/settings	✔	✔	✔

6. Test Users & Demo Credentials

Universal Password

StudentWise123!

Portal Login — /login

Email	Role	Display Name	Linked Record	Best For Demo
sarah@studentwise.test	student	Sarah Ahmed	Student 2026–BSCS–0042	Primary student demo — 6 courses, grades, fees, notifications
hassan@studentwise.test	student	Hassan Raza	Student 2026–BSCS–0043	Pending fees, low attendance scenario
maryam@studentwise.test	student	Maryam Khan	Student 2025–BSEE–0118	EE department, paid fees
teacher@studentwise.test	teacher	Dr. Aamir Khan	Teacher FAC–2018–014	Teacher demo — Reports page

Admin Login — /admin/login

Email	Role	Display Name	Best For Demo
admin@studentwise.test	admin	Portal Admin	Admin CRUD demo — full management panel

Per-User Demo Highlights

Sarah Ahmed (sarah@studentwise.test)

- 6 enrolled courses: CS-304, CS-307, CS-401, CS-403, CS-411, MATH-204
- Real attendance records (e.g., OS at 68% — short attendance alert)
- Fee history + Fall 2026 semester balance (PKR 98,000 total)
- Grades for Fall 2025 and Spring 2025
- 4 notifications (2 unread)

Hassan Raza (hassan@studentwise.test)

- Fee status: Pending
- 3 enrolled courses
- Lower attendance on Operating Systems

Maryam Khan (maryam@studentwise.test)

- Electrical Engineering (BSEE)

- Paid fees, Software Engineering enrollment

Dr. Aamir Khan (teacher@studentwise.test)

- CS faculty, 3 courses
- Access to Reports & Analytics with live charts

Portal Admin (admin@studentwise.test)

- Full Admin Dashboard at </admin/dashboard>
- CRUD students, teachers, courses (persisted to Supabase)

7. Database Schema

Tables (11 total)

Table	Description
departments	CS, EE, BBA, MATH
profiles	User profiles linked to auth.users
students	Student records (ID, GPA, credits, fee status)
teachers	Faculty records
courses	Course catalog with instructor assignment
enrollments	Student ↔ course mapping per semester
attendance_records	Auto-calculated attendance % per enrollment
fee_transactions	Payment history (PKR)
semester_fees	Current semester fee balances
course_grades	Semester grades and grade points
notifications	User-specific alerts (fee, attendance, course, announcement)

Enums

- [fee_status](#): Paid, Pending, Overdue
- [person_status](#): Active, Hold, On Leave

Security

- **RLS enabled** on all public tables
- Students read **own** records only
- Teachers read students, enrollments, attendance (read-only)
- Admins have **full CRUD** on all tables
- Auto-profile trigger on new auth user signup

Migrations

File	Description
20250610000000_initial_schema.sql	Core schema + RLS policies
20250610100000_grades_notifications.sql	Grades, notifications, teacher read policies

Seed Scripts

File / Command	Purpose
supabase/seed.sql	Demo departments, students, courses, enrollments, fees, grades
supabase/seed-auth-users.sql	5 test auth users with fixed UUIDs
npm run seed:users	Admin API seed script (recommended for cloud)

8. Server Functions (API Layer)

All data access goes through TanStack Start server functions in `src/lib/supabase/data.ts`:

Function	Method	Purpose
fetchPortalDashboard	GET	Role-aware dashboard stats and charts
fetchStudentCourses	GET	Enrolled courses with attendance
fetchStudentAttendance	GET	Per-course attendance + summary
fetchStudentFees	GET	Semester fees + transaction history
fetchTeachersDirectory	GET	Faculty directory
fetchStudentResults	GET	Grades grouped by semester
fetchNotifications	GET	User notifications
markNotificationsRead	POST	Mark all notifications as read
fetchReportsData	GET	Analytics for teachers/admins
fetchAdminData	GET	Admin panel data (students, teachers, courses)
fetchDepartments	GET	Department list for forms
adminUpsertStudent	POST	Create/update student
adminDeleteStudent	POST	Delete student
adminUpsertTeacher	POST	Create/update teacher
adminDeleteTeacher	POST	Delete teacher
adminUpsertCourse	POST	Create/update course
adminDeleteCourse	POST	Delete course

Function	Method	Purpose
adminEnrollStudent	POST	Enroll student in course
adminUnenrollStudent	POST	Remove enrollment
adminAssignInstructor	POST	Assign teacher to course

9. Development Timeline & Effort

Git Commit History (June 10, 2026)

#	Commit	Description
1	99f4c4e	Initial commit — StudentWise university dashboard (UI scaffold)
2	f1a1ca4	Initial commit to repo
3	c07795a	Redeploy
4	095cf4d	Vercel account setup and deployment config
5	521a6c1	Removed Lovable branding — custom StudentWise identity
6	5a70dbe	Auth setup — Supabase login, signup, session cookies, test users
7	6850f09	Admin route — dedicated admin login + admin dashboard
8	ee52777	Fix dashboard navigation buttons
9	cdee9f2	Dynamic data — live Supabase integration across all pages
10	529fcaf	Fix 404 errors on console
11	b2d522b	Fix page not found on homepage

Development Effort Summary

Area	Work Completed
UI / UX	10+ portal pages, responsive sidebar layout, admin panel with dialogs/tabs/tables, Recharts dashboards, role-based navigation
Authentication	Email/password login, signup with student linking, SSR cookie sessions, role guards, separate admin login
Database	2 migrations, 11 tables, RLS policies, triggers, seed data for 6 students + 4 teachers + 9 courses
Backend integration	19 server functions, typed Supabase client, admin CRUD mutations
DevOps	Vercel deployment (Nitro preset), env config, cache headers, Node 22 runtime
Documentation	README, AUTH_SETUP.md, user.md, seed scripts

Area	Work Completed
Code volume	262 files changed since initial commit (~109K insertions)

Key Milestones (for PPT slides)

- 1. **Phase 1 — Foundation:** React dashboard UI with shadcn components
- 2. **Phase 2 — Deploy:** Vercel + Nitro serverless + custom branding
- 3. **Phase 3 — Auth:** Supabase Auth, profiles, role-based guards
- 4. **Phase 4 — Admin:** Full admin CRUD panel
- 5. **Phase 5 — Live Data:** Supabase Postgres integration on every page
- 6. **Phase 6 — Polish:** 404 fixes, homepage redirect, production hardening

10. Deployment & Environment

Vercel Configuration (vercel.json)

- Framework: **TanStack Start**
- Build: `npm run build`
- Node.js **22.x** runtime
- Cache-Control: no-store for HTML, immutable for `/assets/**`

Required Environment Variables

Variable	Scope	Required
VITE_SUPABASE_URL	Browser	✓
VITE_SUPABASE_ANON_KEY	Browser	✓
SUPABASE_URL	Server (SSR)	✓
SUPABASE_ANON_KEY	Server (SSR)	✓
SUPABASE_SERVICE_ROLE_KEY	Server (admin seed, signup linking)	✓

NPM Scripts

Command	Description
<code>npm run dev</code>	Dev server at :3000
<code>npm run build</code>	Production build
<code>npm run preview</code>	Preview production build
<code>npm run lint</code>	ESLint
<code>npm run db:push</code>	Push migrations to Supabase
<code>npm run db:reset</code>	Reset local DB with seed
<code>npm run db:types</code>	Regenerate TypeScript types

Command	Description
<code>npm run seed:users</code>	Create test auth users via Admin API

11. Demo Walkthrough (Suggested PPT Flow)

Slide-friendly demo script

1. **Landing** → Open `{BASE_URL}` → auto-redirect to login
2. **Student view** → Login as `sarah@studentwise.test` / `StudentWise123!`
3. **Dashboard** → Show GPA, attendance chart, course progress
4. **Courses** → 6 enrolled courses with attendance badges
5. **Attendance** → Ring charts, short-attendance warning on OS
6. **Fees** → PKR balance, payment history
7. **Results** → Semester grades table, CGPA
8. **Notifications** → Unread fee + attendance alerts
9. **Logout** → Sign out
10. **Teacher view** → Login as `teacher@studentwise.test` → **Reports** page with charts
11. **Admin view** → Go to `/admin/login` → `admin@studentwise.test` → CRUD a student or course live

12. Project Structure

```
studentwise-dash/  
├── src/  
│   ├── routes/                                # File-based pages (TanStack Router)  
│   │   ├── admin/                            # Admin login + dashboard  
│   │   ├── dashboard.tsx                    # Main dashboard  
│   │   ├── courses.tsx                      # Enrolled courses  
│   │   ├── attendance.tsx                   # Attendance tracking  
│   │   ├── fees.tsx                         # Fee management  
│   │   ├── teachers.tsx                     # Faculty directory  
│   │   ├── results.tsx                      # Academic grades  
│   │   ├── reports.tsx                      # Analytics (teacher/admin)  
│   │   ├── notifications.tsx  
│   │   ├── settings.tsx  
│   │   ├── login.tsx / signup.tsx  
│   │   └── index.tsx                        # Root redirect  
│   ├── components/  
│   │   ├── AppLayout.tsx                    # Sidebar + header shell  
│   │   ├── AdminLayout.tsx                  # Admin panel shell  
│   │   └── ui/                              # shadcn/ui components (40+)  
│   ├── lib/  
│   │   ├── supabase/                        # Client, server, auth, data layer  
│   │   ├── auth-guards.ts                   # Route protection  
│   │   └── database.types.ts  
│   ├── hooks/  
│   │   └── use-auth.ts  
└── supabase/
```

```
├── migrations/           # Schema + RLS
├── seed.sql              # Demo data
├── seed-auth-users.sql   # Test accounts
├── scripts/
│   └── seed-test-users.mjs # Admin API user seeder
├── docs/
│   ├── PROJECT_SPEC.md   # This document
│   └── AUTH_SETUP.md     # Auth configuration guide
├── vercel.json
├── nitro.config.ts
└── package.json
```

13. Future Enhancements (Roadmap)

Priority	Feature	Status
High	Settings page — persist profile updates to Supabase	UI only
High	Fee payment gateway integration	Not started
Medium	Teacher — mark attendance for students	Not started
Medium	Real-time notifications via Supabase Realtime	Infrastructure ready
Medium	PDF export for results / fee receipts	Button UI exists
Low	Mobile sidebar drawer	Partial (desktop sidebar)
Low	Email notifications	Toggle in settings (UI only)

14. Quick Reference Card (Copy to PPT)

STUDENTWISE – UNIVERSITY PORTAL

Stack: React 19 · TanStack Start · Supabase · Vercel
Repo: github.com/mangiai/studentwise-dash
Local: <http://localhost:3000>

TEST USERS (password: StudentWise123!)
Student: [sarah@studentwise.test](#) → /login
Teacher: [teacher@studentwise.test](#) → /login
Admin: [admin@studentwise.test](#) → /admin/login

KEY ROUTES
[/dashboard](#) [/courses](#) [/attendance](#) [/fees](#) [/results](#)
[/teachers](#) [/notifications](#) [/reports](#) [/admin/dashboard](#)

End of specification document.